

KANHAIYA BHARDWAJ

Data Scientist | Machine Learning | Python | SQL

Delhi, India | +91 8171879902 | yb13390@gmail.com

linkedin.com/in/kanhaiya-bhardwaj-57788b375 | github.com/kanhaiya668

PROFESSIONAL SUMMARY

Data Science fresher (BCA, 2026) with hands-on experience building 6 end-to-end ML projects — spanning regression, classification, segmentation/recommendation, and time-series forecasting — from data cleaning to deployment as web apps and REST APIs. Strong in Python, SQL and the full ML workflow, with 500+ problems solved on LeetCode. Looking for an entry-level Data Scientist / Data Analyst role.

TECHNICAL SKILLS

Programming: Python, SQL

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI

Machine Learning: Scikit-learn, XGBoost, LightGBM

Deep Learning (basics): TensorFlow, Keras

Web & Deployment: Flask, FastAPI, Streamlit, Docker (basics), AWS (basics), Render

Tools: Git, GitHub, VS Code, Google Colab, Jupyter, Excel, Statistics, EDA

Concepts: Regression, Classification, Clustering, Recommendation Systems, Time-Series Forecasting, Model Explainability (SHAP), NLP (basics), LLM/RAG (basics)

PROJECTS

House Price Prediction — End-to-End ML Project — *Python, Pandas, Scikit-learn, Flask, Docker*

- Built a complete regression pipeline: data cleaning, feature engineering, model training and evaluation on the Kaggle House Prices dataset.
- Compared Linear Regression, Random Forest and Gradient Boosting; best model achieved $R^2 = 0.937$.
- Deployed the model as a Flask web app with a REST API for real-time price predictions.

Customer Churn Prediction — Business Insights — *Python, Pandas, Scikit-learn, Streamlit, Matplotlib*

- Performed EDA, handled missing values and encoded categorical features for a customer churn dataset.
- Trained and tuned Logistic Regression, Random Forest and XGBoost classifiers (~93% accuracy).
- Built an interactive Streamlit dashboard combining churn predictions with business insights.

Credit Card Fraud Detection — *Python, Pandas, Scikit-learn, Custom SMOTE, Flask*

- Handled highly imbalanced transaction data using a custom SMOTE implementation.
- Trained and compared 3 models; best model (HistGradientBoosting) reached 90% precision, 79% recall, F1 0.84.
- Built a Flask app supporting single-transaction checks, bulk CSV upload and a live fraud dashboard.

Customer Segmentation & Recommendation Engine — *Python, Pandas, Scikit-learn (K-Means), Flask*

- Performed RFM (Recency, Frequency, Monetary) analysis on 540K+ UK e-commerce transactions and applied K-Means clustering to segment ~4,300 customers into 4 actionable groups (Champions, Loyal, At Risk, New/Promising).
- Built an item-based collaborative filtering recommendation engine using cosine similarity on purchase history.
- Deployed as a Flask web app (live on Render) returning customer segment and personalized product recommendations by Customer ID lookup.

End-to-End Loan Approval / Risk Prediction — *Python, FastAPI, HistGradientBoosting, SHAP*

- Built a loan-approval classification pipeline on ~4,300 applicant records, identifying CIBIL score as the dominant approval factor via permutation importance.
- Integrated SHAP for per-prediction explainability, surfacing the top contributing factors behind each decision.
- Deployed as a FastAPI app with single prediction, bulk CSV upload, and a live applicant-stats dashboard.

Demand / Sales Forecasting System — *Python, LightGBM, Streamlit*

- Engineered time-series features (lag values, rolling averages, promo/holiday flags) on daily sales data from 1,115 stores (Rossmann Store Sales).
- Trained a LightGBM regression model achieving MAPE 8.32% (RMSE 784.63) on a time-based holdout split.
- Deployed as an interactive Streamlit app (live on Streamlit Community Cloud) for store-level sales forecasting.

EDUCATION

Bachelor of Computer Applications (BCA)

Dr. Bhimrao Ambedkar University | Passing Year: 2026 | 69.35%

ACHIEVEMENTS

- Built 6 end-to-end data science projects, each deployed as a working web app / API.
- Solved 500+ coding problems on LeetCode (Easy & Medium).